

CYBERSECURITY LAB BUILD AND WEB APPLICATION SECURITY TESTING REPORT

Internship Practical Report

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1. Introduction

Cybersecurity testing plays an important role in identifying vulnerabilities and understanding the behaviour of modern web applications. In this practical lab setup, Kali Linux was used as the primary penetration testing operating system within a VirtualBox virtualized environment. Docker was configured to deploy the OWASP Juice Shop vulnerable web application locally for security testing and analysis. The project involved monitoring web traffic using Burp Suite and analyzing network packets using Wireshark. Basic testing activities such as login analysis, search input validation, HTTP request monitoring, and network validation were performed successfully.

2. Objectives

The main objectives of this project are:

- To build a cybersecurity testing lab using Kali Linux and VirtualBox
- To configure Docker inside Kali Linux
- To deploy OWASP Juice Shop locally using Docker
- To validate network connectivity and perform scanning
- To analyze HTTP requests using Burp Suite
- To capture network packets using Wireshark
- To understand basic web application testing concepts

3. Tools and Technologies Used

<i>Technology</i>	<i>Purpose</i>
Kali Linux	Penetration testing operating system
VirtualBox	Virtual machine environment
Docker	Container deployment platform
OWASP Juice Shop	Vulnerable web application
Burp Suite	Web traffic interception and monitoring
Wireshark	Network packet analysis
Nmap	Network scanning and validation
Firefox Browser	Web application access

4. Virtual Lab Setup and Configuration

4.1 Virtual Machine Setup

A Kali Linux virtual machine was created and configured using Oracle VirtualBox. The virtual environment was configured with internet connectivity and full-screen mode support.

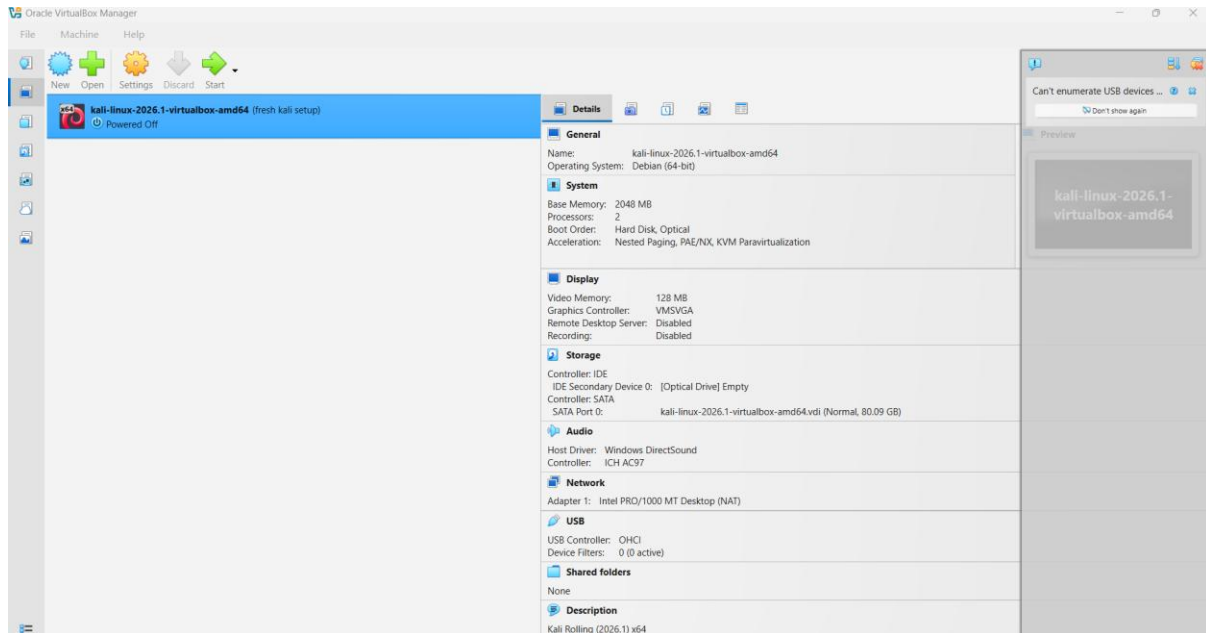


Figure 1: Kali Linux virtual machine configured in VirtualBox.

4.2 IP Configuration

The IP configuration of the Kali Linux virtual machine was verified using the "ip a" command. This confirmed successful network adapter configuration and connectivity.

Command Used:

ip a

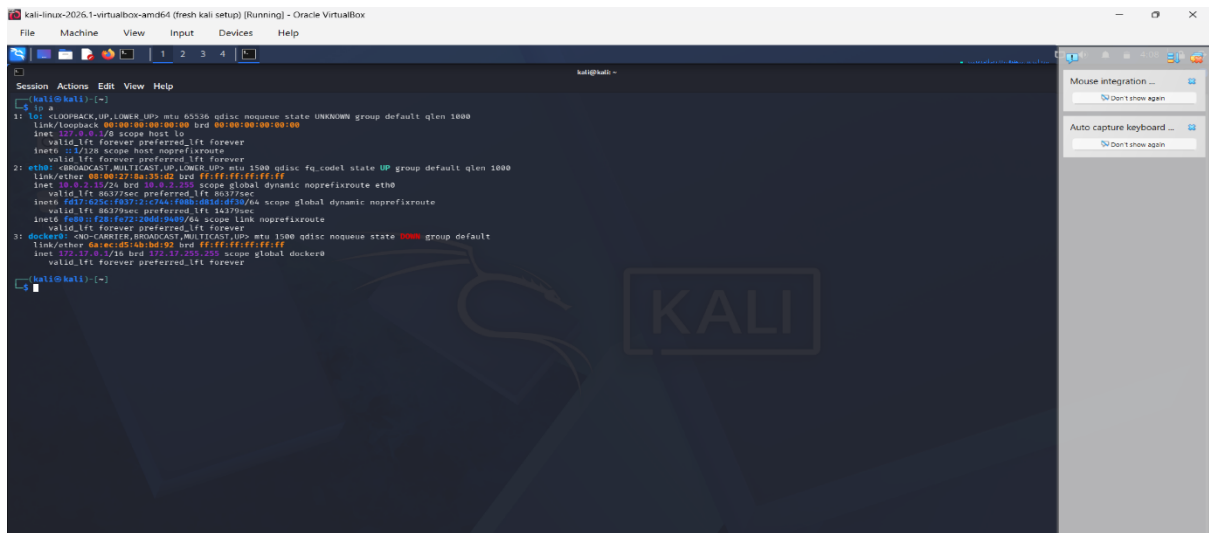


Figure 2: IP configuration of Kali Linux virtual machine.

4.3 Network Connectivity Validation

Internet connectivity was validated using the "ping google.com" command. Successful packet transmission and reception confirmed active internet access within the virtual machine.

Command Used:

ping google.com

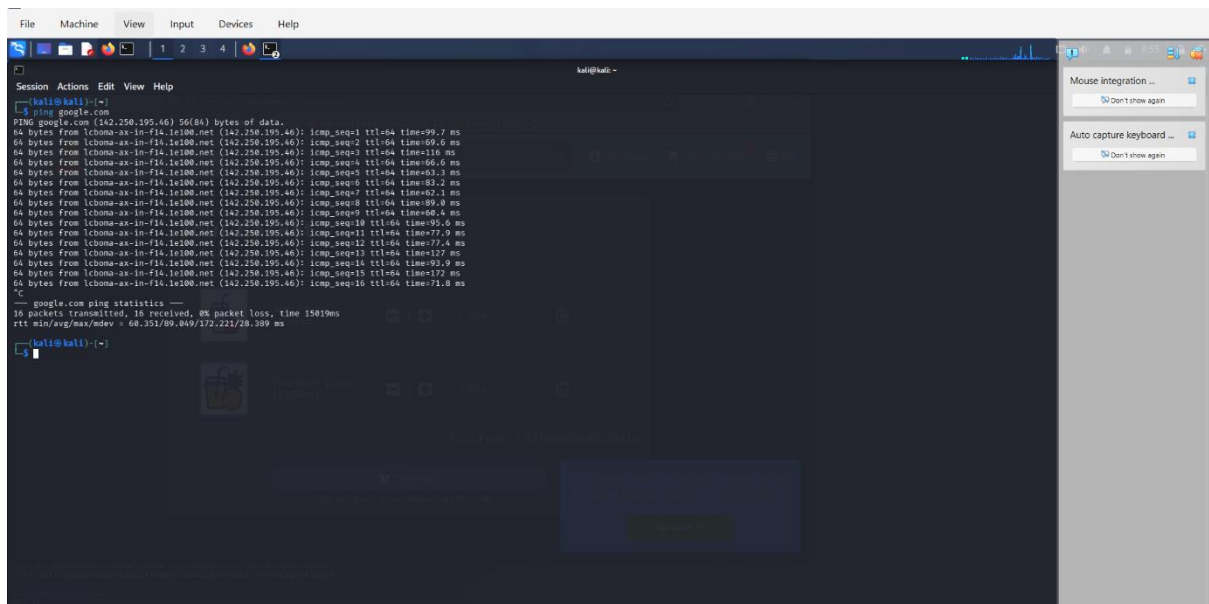


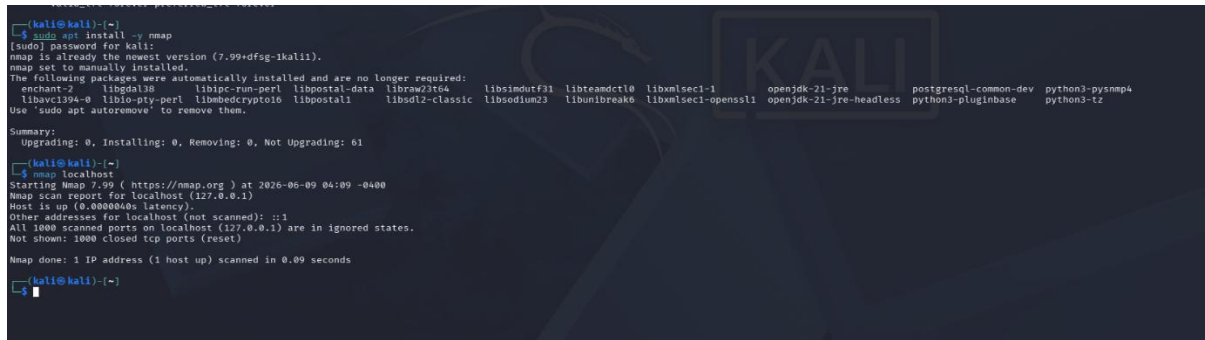
Figure 3: Successful network connectivity validation using ping command

4.4 Nmap Validation

Nmap was installed and used to perform a local host scan. This confirmed that network scanning functionality was operating correctly within the Kali Linux environment.

Command Used:

nmap localhost



```
kali@kali:~$ sudo apt install -y nmap
[sudo] password for kali:
nmap is already the newest version (7.99+dfsg-1kali1).
nmap set to manually installed.
The following packages were automatically installed and are no longer required:
  enchant-2 libgdal30 libipc-run-perl libpostal-data libraw23t64 libsimdutf31 libteamdctl0 libxmlsec1-1 openjdk-21-jre postgresql-common-dev python3-pysnmp4
  libavc1394-0 libio-pty-perl libmbedcrypto10 libpostal1 libstd2-classic libstdc++6 libunibreak libxmlsec1-openssl openjdk-21-jre-headless python3-pluginbase python3-tz
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 61
kali@kali:~$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 04:09 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000000s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
kali@kali:~$
```

Figure 4: Nmap localhost scan results.

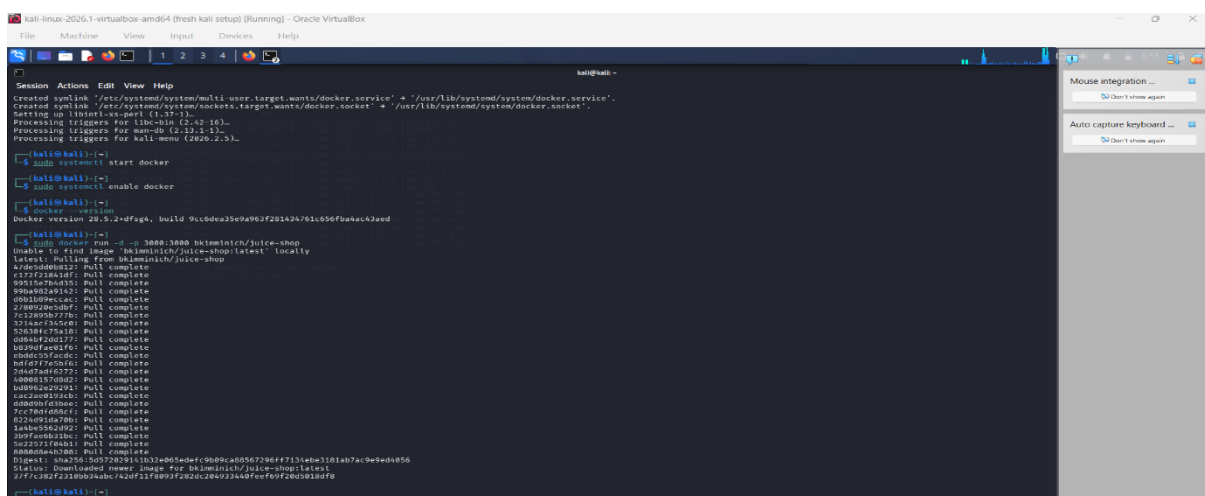
4.5 Docker Installation and Verification

Docker was installed successfully in Kali Linux to support containerized application deployment. Docker version verification and container status were checked using terminal commands.

Commands Used:

docker --version

sudo docker ps



```
kali@kali:~$ sudo apt install -y docker
[sudo] password for kali:
Creating symlink /etc/systemd/system/multi-user.target.wants/docker.service -> /usr/lib/systemd/system/docker.service.
Created symlink '/etc/systemd/system/containers.target.wants/docker.socket' -> /usr/lib/systemd/system/docker.socket.
Setting up libnftnl1:amd64 (1.3.7-1).
Processing triggers for libnftnl1:amd64 (1.3.7-1).
Processing triggers for man-db (2.13.1-1).
Processing triggers for kali-menu (2026.2.2).
kali@kali:~$ sudo systemctl start docker
kali@kali:~$ sudo systemctl enable docker
kali@kali:~$ docker --version
Docker version 28.5.2+dfsg, build 9c16de239e995f221426781c956fbae43aed
kali@kali:~$ sudo docker run -d --name=juice-shop bhimminich/juice-shop
Unable to find image 'bhimminich/juice-shop:latest' locally
latest: Pulling from bhimminich/juice-shop
64ced88a122: Pull complete
c172718a1df: Pull complete
9081a7ba431: Pull complete
908a082a91a2: Pull complete
093a109ec4e1: Pull complete
2780928e308f: Pull complete
7c12a958777b: Pull complete
3714ac7a5e08: Pull complete
52604c75a18: Pull complete
0064df20d177: Pull complete
b239f4e9a145: Pull complete
ebddc55fac8c: Pull complete
b0f0f763b061: Pull complete
704d7ad65772: Pull complete
4880813708a2: Pull complete
b08962e29291: Pull complete
c0c0e9312c26: Pull complete
d0809f63b0e4: Pull complete
7c78af6088c4: Pull complete
622a491da70b: Pull complete
1a48a56c4092: Pull complete
3b9fa08a11bc: Pull complete
5e22271f8a11: Pull complete
88808a8a2081: Pull complete
Digest: sha256:5d7797911b07b065e0e9089ca0567796ff713140e1181ab7ac9e9eda856
Status: Downloaded newer image for bhimminich/juice-shop:latest
27f7c892f2310034b6c7a2f17809f2828c40a23448fcef0af28b093b0f8
kali@kali:~$
```

Figure 5: Docker installation verification.

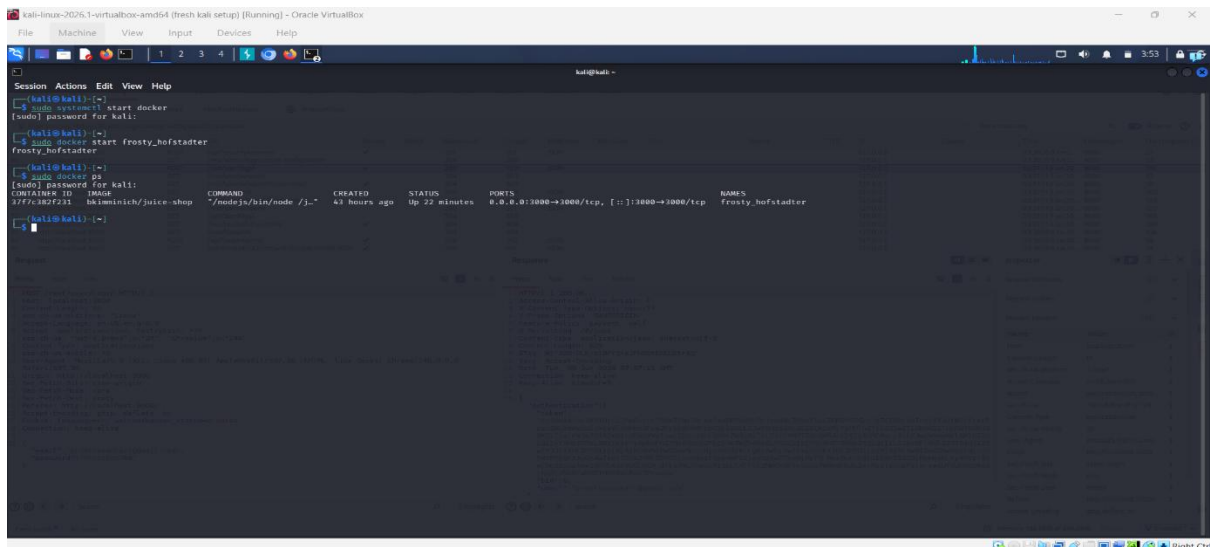


Figure 6: Running Docker container for OWASP Juice Shop.

4.6 OWASP Juice Shop Deployment

OWASP Juice Shop was deployed successfully using Docker and accessed through the browser using localhost.

URL Used:

<http://localhost:3000/>

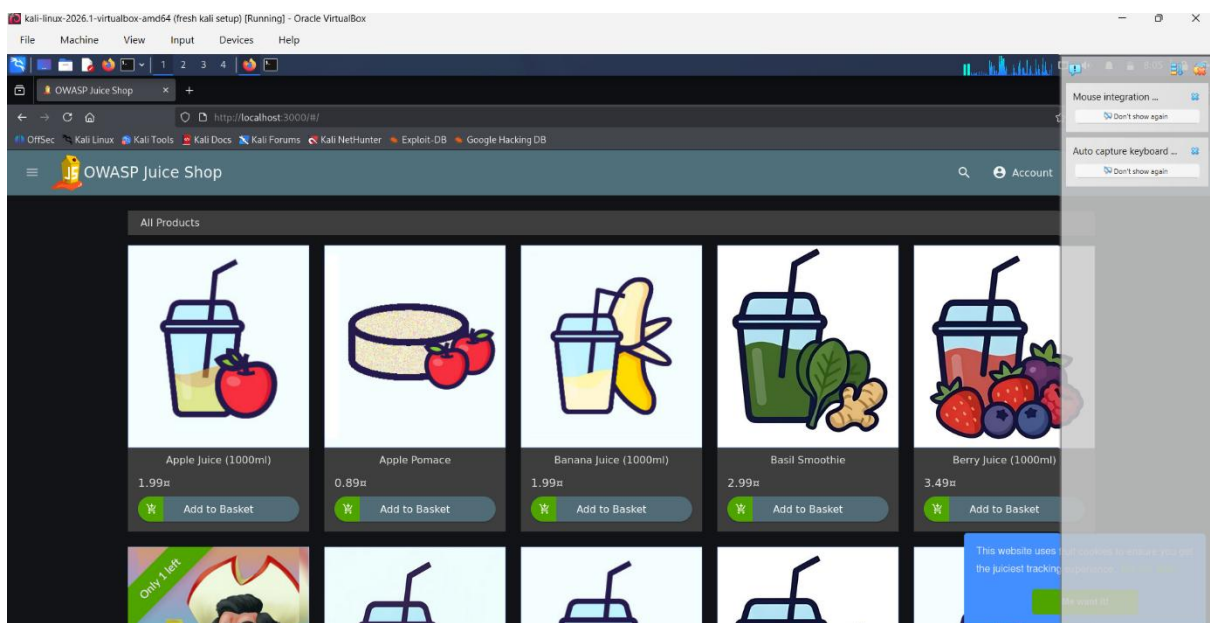


Figure 7: OWASP Juice Shop running successfully in browser.

5. Web Application Testing

5.1 User Authentication Testing

The authentication functionality of OWASP Juice Shop was tested successfully. Login behavior, session handling, and authentication requests were monitored using Burp Suite during application interaction.

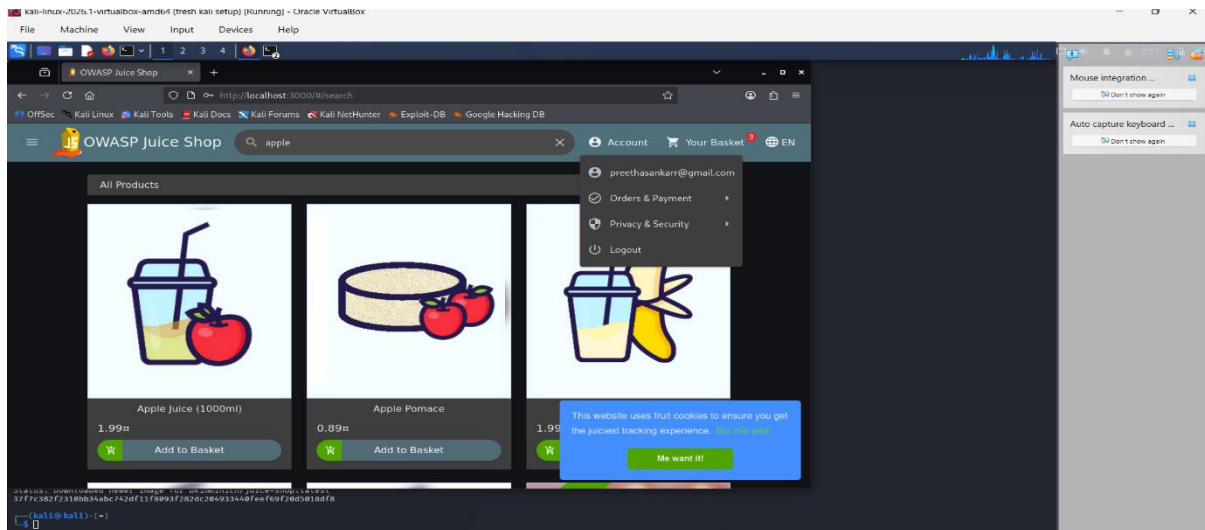


Figure 9: Successful login into OWASP Juice Shop.

5.2 Search and Input Validation Testing

The search functionality of OWASP Juice Shop was tested to observe how the application handles user input and search queries. Different search operations were performed to understand application response behaviour and request handling.

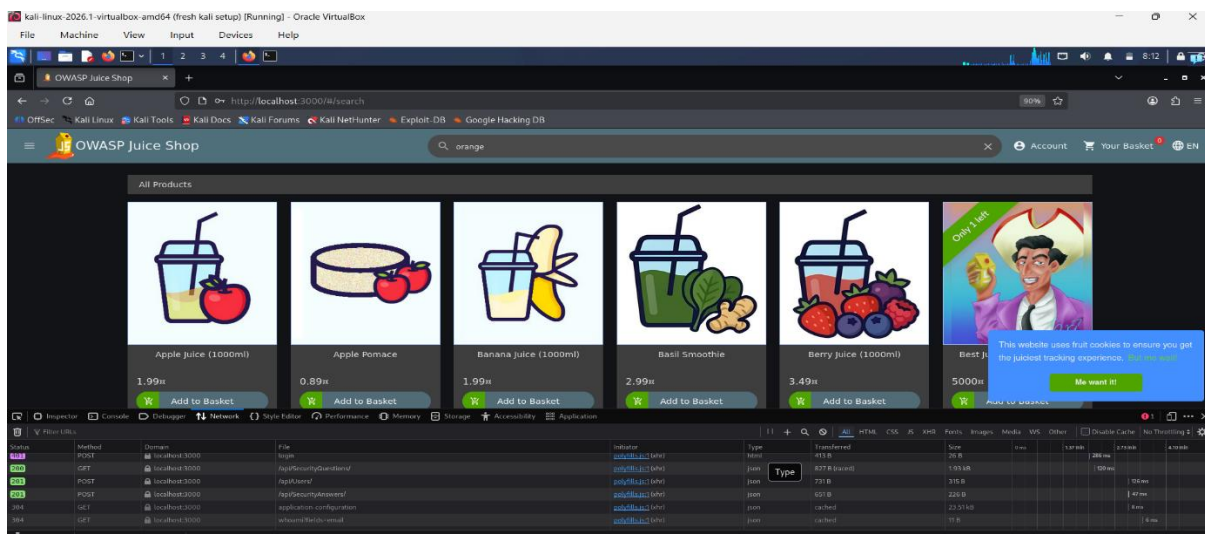


Figure 10: Search functionality testing in OWASP Juice Shop.

5.3 Basket Functionality Testing

The basket functionality of the application was tested by adding and removing products from the shopping cart. This demonstrated interaction with the application's business logic.

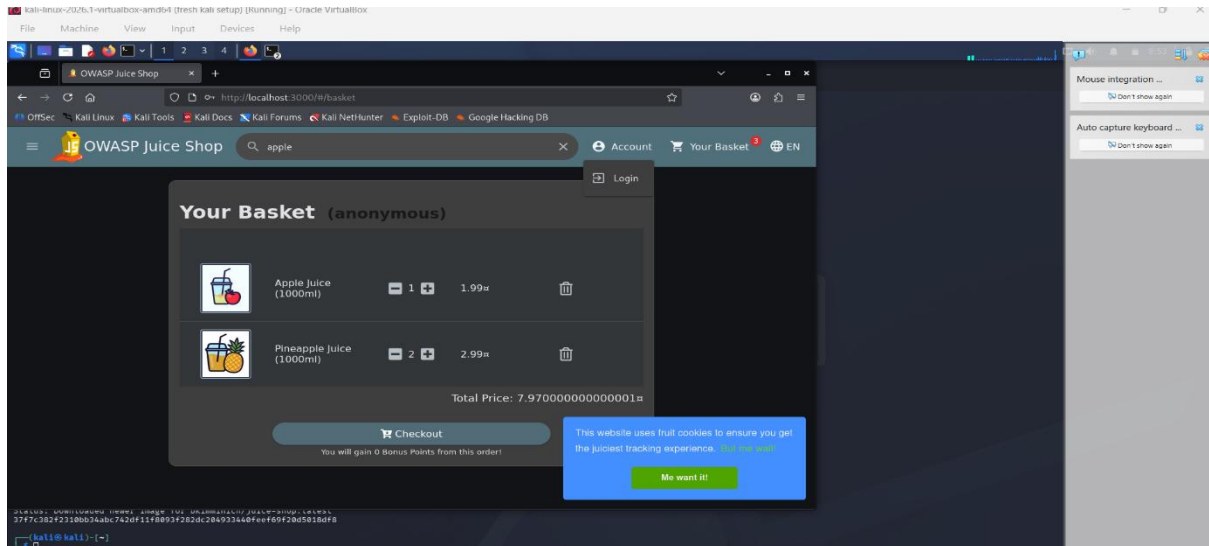


Figure 11: Basket functionality testing.

6. Traffic Monitoring and Analysis

6.1 Burp Suite Traffic Monitoring

Burp Suite was configured successfully to monitor HTTP requests between the browser and OWASP Juice Shop. The HTTP History feature captured requests and responses generated during application interaction.

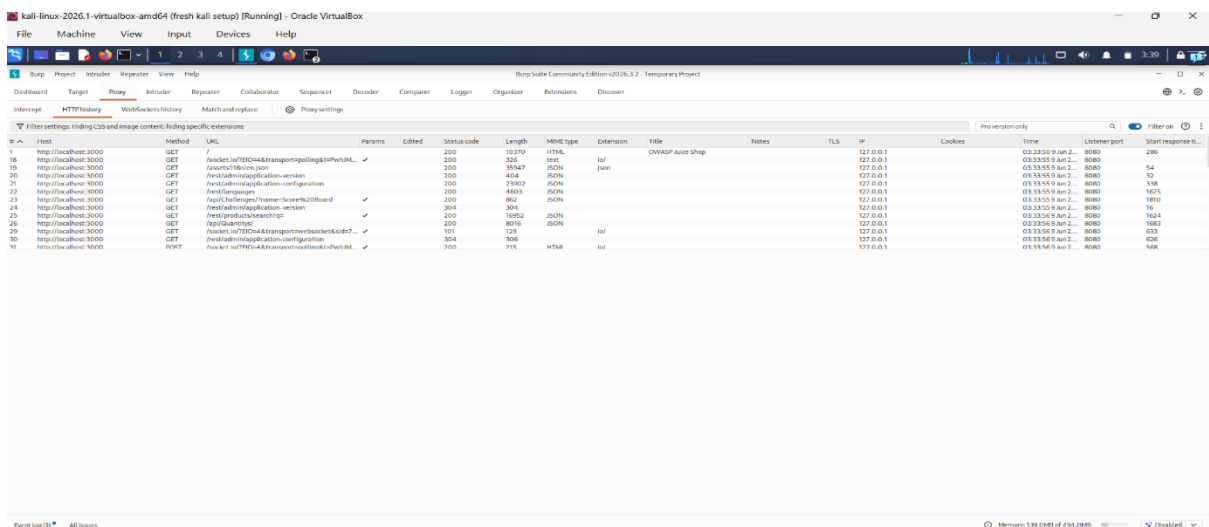


Figure 12: HTTP traffic captured using Burp Suite.

6.2 Authentication Request Analysis

Authentication requests generated during user login were analyzed using Burp Suite. POST requests containing login-related data were captured and inspected.

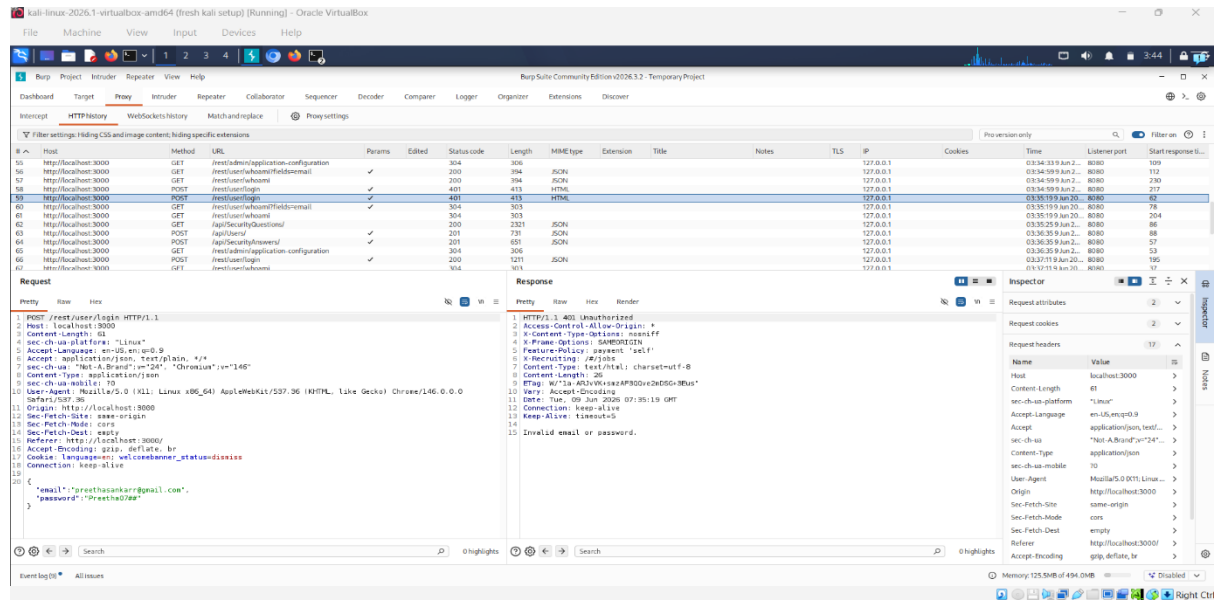


Figure 13: Login POST request captured using Burp Suite.

6.3 Wireshark Packet Capture

Wireshark was installed and used to capture live network packets generated while accessing OWASP Juice Shop. Packet-level traffic analysis was successfully demonstrated.

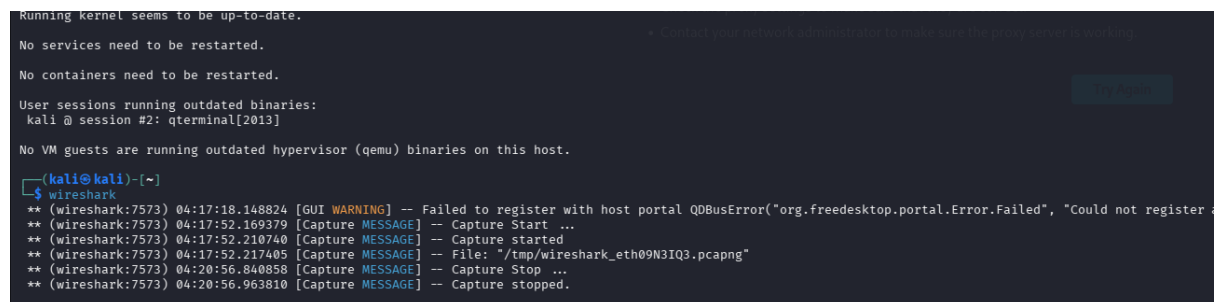


Figure 14: Network packet capture using Wireshark.

7. Observations

- Kali Linux virtual machine was configured successfully.
- Internet connectivity was verified using ping commands.
- Docker was installed and configured correctly.
- OWASP Juice Shop was deployed successfully using Docker.
- Burp Suite successfully intercepted HTTP requests.
- Authentication traffic and HTTP history were analyzed.
- Wireshark successfully captured live network packets.
- Basic web application testing activities were performed successfully.

8. Conclusion

The cybersecurity testing lab was successfully built using Kali Linux and VirtualBox. Docker was configured to deploy OWASP Juice Shop locally for practical web application testing. Burp Suite was used to intercept and monitor HTTP traffic, while Wireshark was used for packet-level network analysis. Basic testing activities including authentication analysis, input validation testing, and traffic monitoring were completed successfully. This project provided practical exposure to cybersecurity tools, web application testing concepts, and network traffic analysis in a controlled lab environment.